MATH 4720/MSSC 5720 Fall 2026 Quiz 2

Name: _____ **Student ID:** _____

1. **Normal distribution**: Head lengths of brushtail possums follow a normal distribution with mean $\mu = 100$ mm and standard deviation $\sigma = 4$ mm, i.e., $N(100, 4^2)$.
   (a) Calculate the z-scores for possums with head lengths of 92 mm and 104 mm.
   (b) Which observation (92 mm or 104 mm) is more unusual or less likely to happen than the other observation? Briefly explain.

Answer:

2. **Binomial Distribution**: Offspring peas have either yellow or green pods. Assume that we want to find the probability that among five offspring peas, exactly two of them have **green** pods. Also, 75% of offspring peas have **yellow** pods. If we model the probability using the binomial distribution, identify the values of

- n, the number of trials,
- x, the number of "successes"
- π, the probability of "success"

No need to calculate any probability.

Answer: